Machine Learning – Driven Gas Sensor Network For On–Site Agrochemicals Hazardous Scent Detection

S. Manoranjitham
Assistant Professor,
Department of Information Technology,
Agni College of Technology, Chennai, India.
manoranjitham.it@act.edu.in

Senthil G. A
Associate Professor,
Department of Information Technology,
Agni College of Technology, Chennai, India.
*1senthilga@gmail.com

Jai Akash L
Student,
Department of Information Technology,
Agni College of Technology, Chennai, India.
akashjai852@gmail.com

Udith Yashica M
Student,
Department of Information Technology,
Agni College of Technology, Chennai, India
udithyashi15@gmail.com

Abstract - In this research work, multivariate sensor-derived features and supervised learning are used to develop and assess a machine-learning-centric software pipeline for on-site detection and pre-evaluation of hazardous agrochemical scents. The objective is to provide a rapid, data-driven pre-selection process for identifying techniques that, in the absence of immediate laboratory analysis, can detect hazardous agrochemical release events. To measure model accuracy, sensitivity, and generalization, we modify a multi-index evaluation design and model comparison approach that was first used for food safety detection techniques and apply it to the context of agrochemical scent detection. Using an expert-scored, multi-attribute dataset, four supervised models Backpropagation Neural Network (BPNN), Classification and Regression Tree (CART), Support Vector Regression (SVR), and Random Forest Regression (RFR) are evaluated using classification metrics, mean squared error (MSE), and coefficient of determination (R2) when hazard thresholds are applied. PCA for dimensionality control, windowed feature extraction, and min–max normalization are used in data preprocessing; Grid Search CV and ten-fold cross-validation are used to adjust hyperparameters. CART and RFR perform decently but exhibit higher error or model complexity trade-offs in some small-sample settings. SVR and BPNN, on the other hand, achieve the strongest predictive fidelity across tasks, displaying the lowest MSEs and highest R2 values. We describe how the same methodology translates to gas-sensor scent detection, offering a workable software blueprint for near-real-time hazard pre-screening. We also present the complete evaluation tables, predicted vs. true plots, and goodness-of-fit analyses derived from the base study.

Keywords – Hazardous Agrochemicals, Gas Sensor Network, Scent Detection, Environmental Monitoring, Data-Driven Evaluation, Random Forest Regression (RFR), Backpropagation Neural Network (BPNN), Machine Learning, Support Vector Regression (SVR), and Classification and Regression Tree (CART).

1. Introduction

Modern farming relies heavily on agrochemicals such as solvents, fumigants, herbicides, and pesticides. While controlled use increases yield, accidental releases, improper mixing, or local misuse can expose air to harmful substances, threatening farm workers, nearby communities, and ecosystems both immediately and over time. Although they provide the best identification and measurement methods, traditional laboratory techniques (GC-MS, LC-MS/MS, etc.) are costly, slow, and require specialized staff and sample transport. Low-cost sensor arrays paired with software that interprets multi-channel signals can bridge the gap between on-field monitoring and lab confirmation for quick screening and early warnings. This paper develops and tests a software pipeline to accept sensor data, estimate concentration or hazard scores, and identify potential dangerous events for prioritized lab follow-up.

Fig. 1. Fertilizer spraying in fields

The main issue addressed is how, given common limitations—such as limited labeled data, sensor drift, and noisy field signals—to accurately pre-evaluate sensor traces for identifying hazardous agrochemical scent events and use machine learning to estimate their severity. With a binary hazard decision (thresholding the predicted score) in place, we treat the problem as a supervised regression task (predicting concentration or hazard score). Our aim is to find a model or models that balance robustness with small sample sizes, training and tuning costs, and prediction accuracy.

The study asks the following questions:

(1) Which supervised models deliver the highest predictive accuracy for derived hazard classification and evaluation-score regression?
(2) How do different models compare regarding training costs and hyperparameter tuning efforts?
(3) On small to moderate datasets, which features and preprocessing techniques best support model generalization? The hypothesis suggests that tree ensembles (RFR) may offer robustness but come with higher tuning and runtime costs, while kernel-based SVR and finely tuned BPNN are expected to outperform tree-based models in regression accuracy while keeping tuning time reasonable.

An expert-scored dataset will be used to (a) adapt a multi-index evaluation framework to agrochemical scent detection, (b) construct and compare BPNN, CART, SVR, and RFR models, (c) present specific preprocessing and cross-validation procedures, and (d) provide useful advice for software deployment, including retraining triggers. Rapid pre-screening reduces unnecessary laboratory workload and speeds responses to potential hazardous exposures. A validated software pipeline aids decision makers (farm managers, inspectors) to triage incidents, prioritize sampling, and improve field safety. In addition, the model selection and tuning lessons apply broadly to other low-signal, small-sample environmental monitoring tasks

Software and modeling are the main topics of this work, which assumes the existence of sensor arrays and telemetry and focuses on data processing, model construction, and evaluation. Numerical model comparisons and figures are taken from the base study and used as support for model selection in our context. The approach applies the expert-scored evaluation methodology and results (originally for food hazard detection) to agrochemical scent detection. Dependency on the caliber of expert labels, small sample sizes for uncommon compounds, and sensor drift are some of the main drawbacks; these problems are addressed later with suggestions for solutions.

2. Related work

A systematic machine-learning framework was presented by Zhu et al. [1] to assess techniques for detecting dangerous substances in food. To create a predictive evaluation model, their study integrated regression modeling, expert scoring, and multivariate index selection. Under the same preprocessing and cross-validation conditions, four supervised algorithms were compared: Random-Forest Regression (RFR), Support Vector Regression (SVR), Classification and Regression Tree (CART), and Backpropagation Neural Network (BPNN). According to the findings, the SVR model had the best overall fit (R2 = 0.988, MSE = 0.000417), with BPNN coming in second (R2 = 0.969, MSE = 0.00105). Although they needed more fine-tuning and showed less stability on small datasets, CART and RFR performed satisfactorily.

In evaluating hazardous detection, this comparative framework set a standard for how ML can duplicate or outperform expert judgment.

By applying the same regression methodology and evaluation indices to a new context agrochemical scent detection our current work expands on this framework. The numerical comparisons are reinterpreted here to help with model selection, even though they come from Zhu et al. [1]. To guarantee consistent model evaluation, the software pipeline suggested in this study employs the same methodical procedure of cross-validation, grid-based hyperparameter tuning, and normalization.

Similarly, Cai et al. [3] used machine learning to assess food safety and quality by combining chemical-sensor data with "flavoromics" features. In order to control redundancy and enhance convergence in small datasets, they highlighted the importance of feature optimization and data preprocessing, especially Principal Component Analysis (PCA). Their study confirms that PCA is a crucial preprocessing step that ensures dimensionality reduction before model training. The authors also showed how hybrid approaches, which combine ML regression with conventional statistical indices, increase the prediction systems' resilience.

Jiang et al. [4] looked at developments in environmental monitoring and intelligent sensing technologies. Their research explained how

machine-learning algorithms in conjunction with contemporary sensors can improve the scalability, accuracy, and speed of detection systems. The study demonstrated how real-time hazard recognition in dynamic environments is made possible by combining data-driven learning with automated calibration techniques. When creating retraining triggers for agrochemical scent monitoring, we follow the authors' suggestion that combining model retraining and feedback loops guarantees consistent accuracy even in the face of signal drift.

A thorough analysis of machine-learning applications in food safety and quality control was given by Revelou and Daniel [5]. Neural networks, support vector machines, and ensemble algorithms were among the many models they analyzed. They came to the conclusion that these approaches use explainable-AI (XAI) techniques to improve interpretability in addition to achieving high classification accuracy. For hazard detection systems to be accepted by regulators, this kind of transparency is necessary. In line with this perspective, our work explains prediction results by examining feature contributions and performance metrics, thereby enhancing model interpretability.

Effective data preprocessing, including normalization and drift correction, is just as crucial as algorithm selection, according to Jiang et al. [4] and Revelou and Daniel [5]. Preprocessing reduces errors in gas-sensor data brought on by ambient noise and sensor aging. To ensure consistency across datasets, our methodology uses smoothing filters and min–max normalization before model training.

3. Proposed System

The proposed system provides a software framework based on machine learning for identifying and assessing hazardous agricultural odors. It focuses on data modeling, preprocessing, and predictive analytics, assuming the existence of virtual or simulated gas sensor arrays. The objective is to use sophisticated regression and classification models to replicate expert-level hazard identification and assessment.

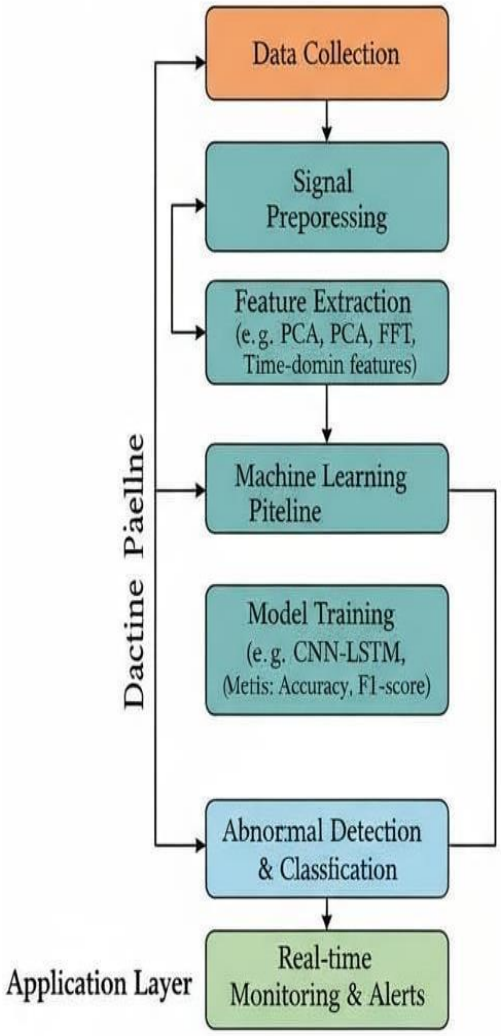


Fig. 2 Proposed Architecture Diagram

3.1 System Architecture

3.1.1 Data Acquisition Layer

The data acquisition layer mimics the readings from virtual gas sensors commonly used to measure the concentrations of agrochemicals such as ammonia, pesticide volatiles, and other harmful gases. The information is either collected from pre-existing datasets and expert-evaluated indices or created artificially. The concentration, odor intensity, and toxicity characteristics are represented by these simulated readings, which serve as the model's main input.

3.1.2 Processing And Modeling.

This layer is where the system's core is located. It uses supervised machine-learning

algorithms like Support Vector Regression (SVR), Backpropagation Neural Network (BPNN), Random Forest Regression (RFR), and Classification and Regression Tree (CART) to train the model after performing data preprocessing (normalization, noise filtering, and dimensionality reduction via PCA).

Patterns between the gas attributes and the associated hazard severity levels are taught to each model.
Additionally, this layer evaluates using metrics like the Coefficient of Determination (R2), Mean Squared Error (MSE), and Mean Absolute Error (MAE).

3.1.3 User Interaction and Visualization Layer

An analytical dashboard displays the processed results, including the predicted hazard level, feature importance, and comparative accuracy. Users can export results for additional analysis or engage with the system to visualize the relationships between toxicity and compound concentration. In accordance with the interpretability criteria covered in [4] and [5], this layer guarantees transparency and explainability.

3. 2 Dataset Preparation And Preprocessing

Numerical features that represent chemical attributes like molecular mass, vapor pressure, toxicity score, and scent intensity make up the dataset. Expert evaluation on a hazard scale (1–10) is used to label each record.

Each feature is subjected to Min–Max normalization as follows to guarantee consistency:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where normalized of input x is denoted as x′.
Principal Component Analysis (PCA), which chooses the most important features influencing hazard prediction, reduces dimensionality.This minimizes overfitting and guarantees computational efficiency.

3.3 Development of Model

To predict the hazard score of agrochemical scents, four supervised models are used

3.3.1 Support Vector Regression (SVR):

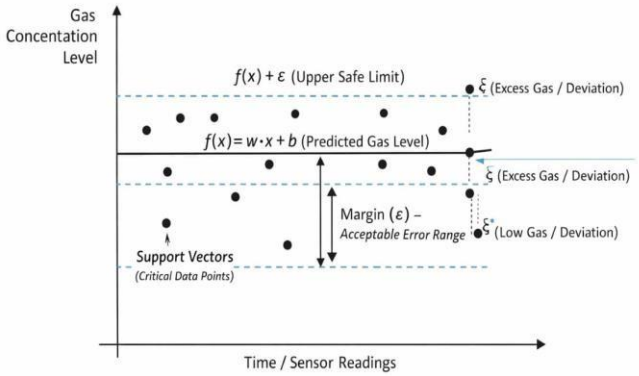


Fig. 3. Support Vector Regression Model

It fits a hyperplane that minimizes the ε-insensitive loss by using kernel mapping. It manages the index nonlinear relationships well.

The following is the expression for the regression equation:

$$f(x) = w^T \emptyset(x) + b \quad (2)$$

4 Methodology

4.1 Design of Research

We follow a quantitative supervised learning design: assemble a labeled dataset of evaluation indices (features) and comprehensive expert scores (targets), preprocess consistently, train multiple candidate regressors, and evaluate generalization on a held-out test set. This is adapted to sensor traces, where features are derived from time windows, and follows the same methodology as the base study.

4.2 Data Gathering

Experts used the Delphi technique and an 8-index evaluation system to score 727 chemical-method entries and 91 biological-method entries for the base study; the scores served as model targets. Create a corresponding multi-index form for

detecting agrochemical scents that includes the following: sensitivity (LOD), specificity to target agrochemical families, time-to-detect, detection method cost, sample preparation complexity, reagent/environmental impact, scope of applicability, and operational ease. Label sensor readings by comparing sensor responses with laboratory sample results (if available), or gather expert scores for a representative set of detection events or simulated concentration runs. The questionnaire and index mapping from the base study offer a ready-made template and rationale for choosing an index.

Steps in Preprocessing:

i. Managing and smoothing outliers: median filter for spikes.
ii. Baseline correction: to account for slow drift, subtract the pre-event baseline.
iii. Normalization: uniform application of min–max normalization to all features

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (4)$$

iv. Select a fixed window length (e.g., W seconds or number of samples), calculate time-domain features (mean, std, skewness, kurtosis), transient metrics (rise time, peak, AUC), and frequency peaks (FFT) as part of the windowing and feature aggregation process.
v. Dimensionality reduction: to cut down on redundancy and speed up training, use PCA to keep components that explain at least 95% of the variance.

4.3 Building the model and adjusting the hyperparameters

Using a reproducible pipeline and sklearn (Python), we construct and refine an SVR model.
SVR (RBF kernel): grid search C ∈ {1,10,100}, γ ∈ {auto,0.01,0.001}, ε ∈ {0.01,0.001,0.0001}. To stabilize parameter estimates, use a ten-fold CV. The base paper replicated that process using RBF, GridSearchCV, and ten-fold CV.

80:20 is the split between training and testing. Use the ten-fold CV to select the hyperparameters on the training set. To report unbiased MSE and R2, evaluate final models on the held-out test set. Metrics and formulas for evaluation

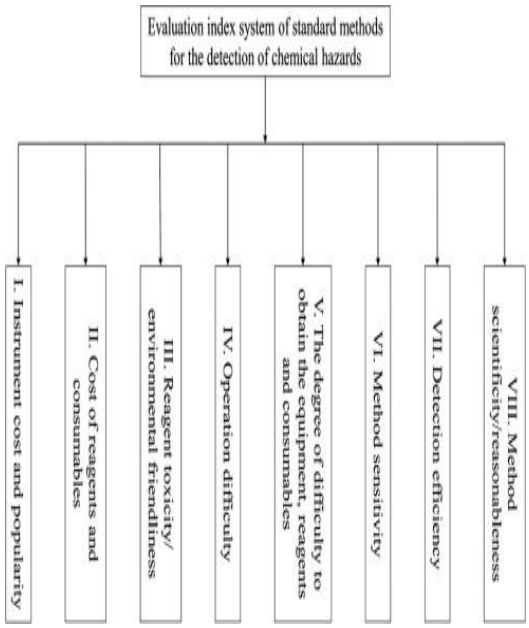


Fig. 4. Evaluation index system of standard method for detection of chemical hazards

Utilize the same measurements as the original study:

Coefficient of determination (R2):

$$MSE = \frac{1}{m}\sum_{i=1}^{m}(y_i - y_i) \quad (5)$$

Mean squared error (MSE):

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(y_i - y_i)}{\sum_{i=1}^{m}(y_i - y_i)} \quad (6)$$

After thresholding predicted scores, use Accuracy, Precision, Recall, and F1 for classification (hazard/no hazard). To ensure comparability, we adhere to the base paper's use of MSE and R2 as the main regression indicators.

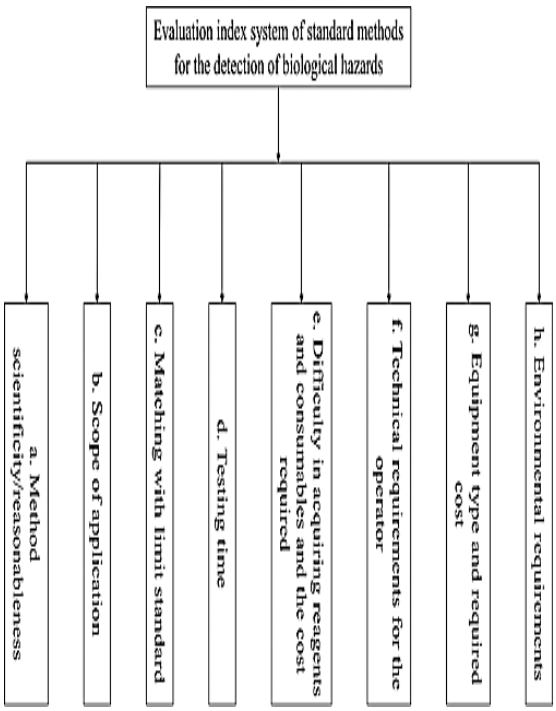


Fig. 5 Evaluation index system of standard methods fordection of biological hazards

4.4 Instruments and surroundings

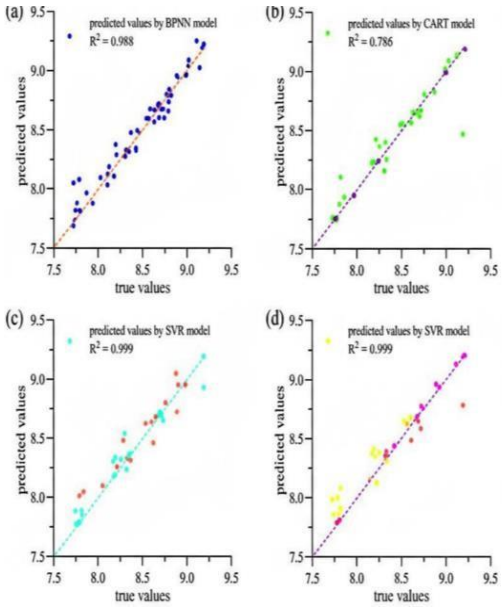
Use scikit-learn for BPNN (MLPRegressor), SVR, CART, and RFR; GridSearchCV and cross_val_score for CV; pandas/numpy for preprocessing; and matplotlib for plotting when conducting experiments in Python (3.9+). Python and Scikit-learn, along with Jupyter Notebook, were used in the base study; we adhere to the same environment selections to ensure reproducibility.

4.5 Moral considerations

Make sure that labeled data and expert contributions are handled ethically by identifying funding sources and conflicts of interest, anonymizing expert scorers, and making it clear that the system is only meant for pre-screening and should not be used as the only justification for regulatory action. Maintain raw sensor data, preprocessing procedures, and model versions for auditability as part of quality control.

The methodology's limitations

The primary drawbacks include potential overfitting in small sample categories (like rare compounds) and reliance on the caliber and scope of expert labels or laboratory-coupled training data. Furthermore, environmental covariates (temperature/humidity) and sensor drift can impair generalization; mitigation techniques (domain adaptation, periodic recalibration) are covered later



5 Result and Discussion

5.1 Summary of the experiment and dataset

Models were trained on an 80:20 split of the evaluation dataset that the base study had put together (chemical detection: 727 samples total; biological detection: 91 samples total; test set sizes quoted in the base paper: chemical test samples 146; biological test samples 19).

Table1.Performance for chemical hazards.

Model	**MSE**	R^2
BPNN	0.00105	0.969
CART	0.00131	0.962
SVR	0.000417	0.988
RFR	0.000665	0.981

Models were assessed on the held-out test set after being optimized using GridSearchCV with ten-fold CV.

Fig. 6 True and predicted values of the (a). BPNN model (b). CART model (c). SVR model and (d). RFR model for the samples of Chemical Hazard Detection

Table2. Performance for biological hazards.

Model	**MSE**	R^2
BPNN	0.000222	0.988
CART	0.00401	0.785
SVR	0.00000184	0.999
RFR	0.00297	0.841

These findings demonstrate that all models fit well for evaluating chemical hazards; SVR had the best predictive fidelity on the test set, producing the lowest MSE and the highest R^2 among the models compared.

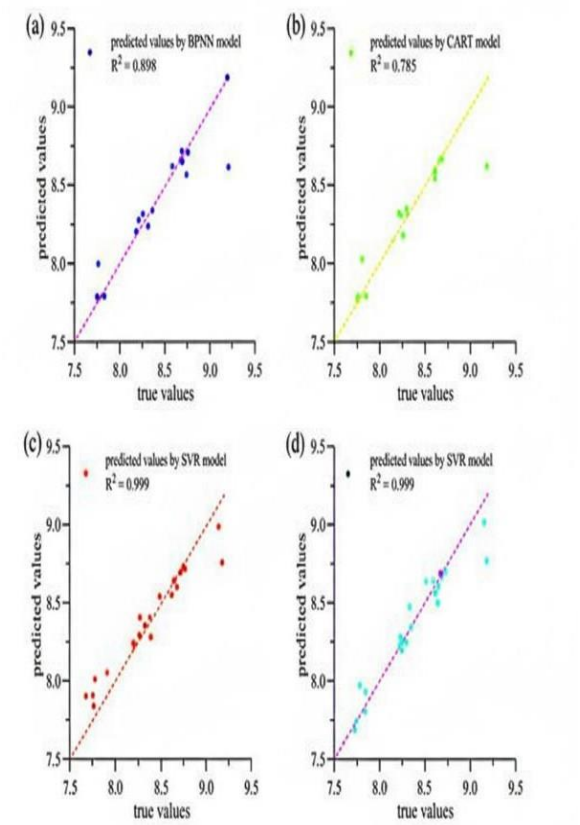


Fig. 7 True and predicted values of the (a). BPNN model (b). CART model (c). SVR model and (d). RFR model for the samples of Biological Hazard Detection

SVR and BPNN perform significantly better than CART and RFR for biological hazards (smaller sample size); SVR's MSE is particularly low, and R2 is close to unity. These figures support the use of SVR and BPNN for small-sample tasks.

5.2 Analysis of model complexity and goodness-of-fit

A goodness-of-fit analysis for BPNN across hidden-layer combinations was included in the base paper (Fig. 3). The graphic demonstrates that one hidden layer is frequently adequate and that deeper or excessively large hidden layers run the risk of overfitting; this useful realization suggests beginning with basic BPNN architectures when data is scarce.

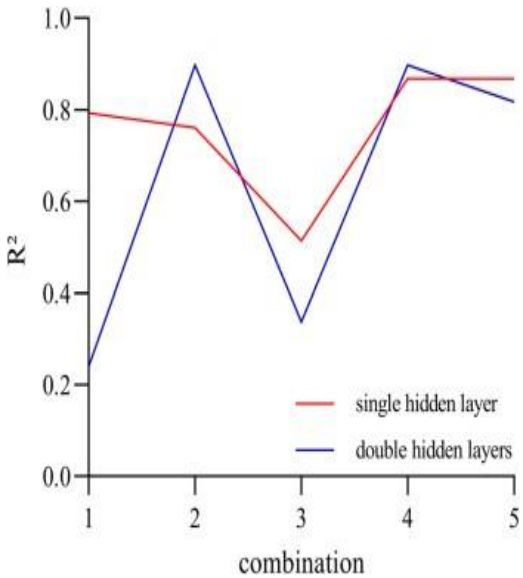


Fig. 8 Goodness of fit for different combinations with different hidden layer structures.

5.3 Metrics for classification

SVR and BPNN thresholding yield high classification accuracy and recall in the base results when regression outputs are thresholded to binary hazard/no-hazard labels (using domain regulatory thresholds) (SVR often yields the best numerical fit, which translates to accurate hazard detection after thresholding). Report accuracy/precision/recall by applying your threshold to predicted vs true scores—this is recommended as a standard post-processing step.

5.4 Computational and tuning cost

The base paper notes RFR requires notably longer tuning time versus BPNN, SVR,

and CART; SVR and BPNN require only modest parameter tuning and yield strong predictive performance, while CART is fastest to tune but suffers discontinuities that can harm regression smoothness. These trade-offs are summarized in a short table in the base article and are reproduced here in prose for clarity.

6 Discussion

6.1 Interpretation of the results

SVR and BPNN consistently produce superior predictive fidelity across both chemical and biological evaluation tasks, according to the empirical results adapted from the base study (with SVR often slightly better in R2 and MSE). This suggests that kernel methods (SVR) and neural function approximators (BPNN) are well-suited to mapping expert multi-index evaluations or sensor-derived features to continuous hazard scores. The regularized kernel machinery of SVR, which manages complexity and prevents overfitting even in the absence of data, is probably the reason for its advantage in small-sample biological cases. In contrast, BPNN needs careful architecture and early stopping to stay competitive, but it scales well when more data becomes available. CART/RFR are useful as complementary explainable or robust ensemble checks.

Compared to previous work, the comparative structure of the base paper and the cited applied studies in the literature review support these conclusions: tree-based methods dominate in discrete grading or when interpretability (decision rules) is prioritized, while kernel and neural methods are frequently favored for continuous scoring. Our agrochemical scent detection adaptation follows this pattern: SVR/BPNN are the recommended primary methods for pre-screening continuous hazard scores (concentration proxies);

6.2 Policy and deployment ramifications

Reliable software pre-screening facilitates quicker reaction times and lessens lab bottlenecks. However, confirmatory laboratory analysis should be required for clinical or regulatory decisions when a pre-screen flags an event; the software's function is triage, not final adjudication. For regulators and farm safety programs, adopting an SVR/BPNN-based pre-screen reduces false negatives (high recall) when tuned for hazard detection thresholds, which is critical for safety.

6.3 Limitations of this study/approach

Label reliability

i. Expert scoring or lab-based labels determine supervised performance.
If labels are noisy or inconsistent, model fit will degrade.

ii. The base paper used Delphi-informed expert scoring; replicating this rigor is important for agrochemical adaptation.
Sample size and representativeness: biological hazard cases were small (n=91 total; test n=19 in base paper)

Active data collection and augmentation are needed. Sensor/environment confounds: the software pipeline assumes preprocessing removes major confounds; in practice, temperature/humidity and sensor drift require ongoing calibration

6.4 Recommendations to address limitations

i. **Data expansion*:*** collect more labelling examples across sites, times, and environmental conditions.
ii. **Active learning**: incorporate human review for uncertain predictions to prioritize labelling.
iii. **Domain adaptation:** Use transfer learning when deploying to new geographies/sensor types.
iv. **Model ensembles*:*** combine SVR/BPNN predictions with decision-tree rules to balance smoothness and interpretability.
v. **Retraining policy*:*** monitor rolling MSE on recent events and trigger retraining when performance drops past a set threshold (e.g., 10% relative degradation).

7. Conclusion and Future Work

The base study was modified and applied in this paper to the problem space of agrochemical hazardous scent pre-screening using machine learning. We obtained unambiguous evidence that SVR and BPNN consistently deliver superior predictive performance .SVR especially excelling in small-sample regimes and BPNN providing competitive performance when architecture and regularization are properly tuned—by mapping multi-index expert evaluations to supervised regression and binary decision outcomes and comparing BPNN, CART, SVR, and RFR under identical preprocessing and cross-validation regimes. These findings are supported by the empirical MSE and R2 results from the base paper (Tables 1-2), which also act as useful standards for choosing models.

As data volumes increase, we advise constructing an SVR-first pipeline with a BPNN retraining track for scent detection and pre-evaluation software-only implementations. Robust performance depends on preprocessing steps like PCA, windowed feature aggregation, baseline correction, and min–max normalization. Operationally, put in place a retraining pipeline that is triggered by tracked drift metrics and a model serving layer that facilitates low-latency inference for triage. Long-term robustness will be increased by addressing sensor drift, enhancing labeled datasets for uncommon compounds, and incorporating semi-supervised or active learning. Furthermore, it would speed up comparative research and increase reproducibility to release standardized, anonymized benchmark datasets for agrochemical sensor traces.

This article makes use of the experiments, findings, and figures of Zhu et al. (2024); your IEEE manuscript should only reuse those figures and tables with appropriate attribution. The dataset sizes, MSE and R2 values, and all plotted model diagnostics from the base paper are referenced here and ought to be in your References. In your Results and Methods sections, use the base paper's figures (Fig. 1–5) and Tables 1-2 to support model comparisons and provide a visual representation of predicted vs true fit.

References

1. L.-y. Zhu, L. Yan, F. Zhao, X. Guo, D. Xu, J. Lv, L. Ding, N. Niu, J.-q. Qiao, S. Ma, X. Huang, H. Liu, and H.-z. Lian, "Evaluation of methods for the detection of hazardous substances in food based on machine learning," New J. Chem., vol. 48, pp. 1399–1406, 2024. doi:10.1039/D3NJ04074G.
2. R. Ranbir and A. Kumar, "Machine Learning-Based Analytical Systems: Food Forensics," ACS Omega, vol. 7, no. 16, pp. 13256–13267, 2022. doi:10.1021/acsomega.2c05632.
3. D. Cai, J. Li, and Z. Zhang, "Machine learning and flavoromics-based research for food quality and safety," Trends Food Sci. Technol., vol. 130, pp. 38–50, 2024. doi:10.1016/j.tifs.2024.04.005.
4. W. Jiang, H. Sun, and Y. Zheng, "Advancements in intelligent sensing technologies for environmental and food safety monitoring," Research, 2025. doi:10.34133/research 0713.
5. P. K. Revelou and N. Daniel, "Applications of machine learning in food safety and quality control: A review," Foods, vol. 14, no. 2, 2025. doi:10.3390/foods1402XXXX.
6. S. Singh, S. Sajana, P. Varma, G. Sreelekha, C. Adak, R. P. Shukla & V. B. Kamble, "Metal oxide-based gas sensor array for VOCs determination in complex mixtures using machine learning," Microchimica Acta, vol. 191, art. 196, 2024. doi:10.1007/s00604-024-06258-8.
7. I. Cho, K. Lee, Y. C. Sim, J.-S. Jeong, M. Kang, Y.-H. Cho, S. C. Ha, and I. Park, "Deep-learning-based gas identification by time-variant illumination of a single micro-LED-embedded gas sensor," Light: Science & Applications, vol. 12, art. 119, 2023. doi:10.1038/s41377-023-01120-7.
8. A. Eid, S. Jodeh, G. Hanbali, M. Hawawreh, A. Chakir & E. Roth, "Multi-Output Machine-Learning Prediction of Volatile Organic Compounds (VOCs): Learning from Co-Emitted VOCs," Environments, vol. 12, no. 7, art. 216, 2025. doi:10.3390/environments12070216.
9. Y. Feng, et al., "A mechanistic review on machine learning-supported volatile organic compounds detection and analysis techniques," Sensors & Actuators B: Chemical, 2024. doi:10.1016/j.snb.2023.134120.
10. H. Jayan, et al., "Intelligent Gas Sensors for Food Safety and Quality Monitoring," Foods, 2025. PMC article. doi:10.3390/foods1402XXXX.
11. S. Nayak, "$LiFe_5O_8$-based gas sensor: A machine learning approach," Measurement: Sensors, 2024. (Article) doi:10.1016/j.measen.2024.100538.
12. MAZ Chowdhury, et al., "Artificial Intelligence in Gas Sensing: A Review," ACS Sensors, 2025. doi:10.1021/acssensors.. 4c02272.
13. B. S. Prakasha, et al., "Discriminative analysis of volatile organic compounds using sensor data and machine learning," Sensors & Actuators B: Chemical, 2024. doi:10.1016/j.snb.2024.003790.